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User defined network access that supports address rotation

Abstract

Methods are provided that support media access control (MAC) address rotation (RCM) by generating a passcode for associating a user defined network by one or more endpoint devices instead of using MAC addresses for their respective device identity. In these methods, a computing device obtains a registration request for establishing a user defined network (UDN) and generates a unique UDN identifier and a unique passcode associated with the unique UDN identifier. The unique passcode enables an authentication of one or more endpoint devices to connect to the UDN. The authentication is independent of the MAC address of a respective endpoint device. The computing device provides the UDN identifier and the unique passcode such that the UDN identifier and the unique passcode are for connecting the one or more endpoint devices to the UDN.

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Background/Summary

TECHNICAL FIELD
(1) The present disclosure generally relates to data and communication networks.
BACKGROUND
(2) When sharing a network by multiple users, network segmentation may be used for security and enhanced control of the user devices. For example, a User Defined Network (UDN) allows the users to create their own personal network (a network segment) that would include only their devices. The user may invite other trusted users into their personal network, sometimes referred to as a private room. This provides the user with security and ability to control the sharing of their devices. The user registers their devices to the private room by inputting a Media Access Control (MAC) address of the respective device, which is then associated with the personal network. In the UDN, communication is allowed only between the registered devices.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a block diagram illustrating a system configured to authenticate one or more endpoint devices onto the UDN using a unique passcode, according to an example embodiment.
(2) FIG. **2** is a flow diagram illustrating a UDN registration process, according to an example embodiment.
(3) FIG. **3** is a flow diagram illustrating a process of onboarding an endpoint device onto a UDN network, according to an example embodiment.
(4) FIG. **4** is a flow diagram illustrating a process of updating UDN information at a UDN client portal based on a MAC address rotation, according to an example embodiment.
(5) FIG. **5** is a flow diagram illustrating a method of generating a unique passcode that is used instead of a media access control (MAC) address to connect an endpoint device to the UDN, according to an example embodiment.
(6) FIG. **6** is a flow diagram illustrating a method of connecting an endpoint device to the UDN based on a passcode provided by the endpoint device, according to an example embodiment.
(7) FIG. **7** is a hardware block diagram of a computing device that may perform functions associated with any combination of operations in connection with the techniques depicted and described in FIGS. **1-6**, according to various example embodiments.
DESCRIPTION OF EXAMPLE EMBODIMENTS
(8) Overview
(9) A mechanism is presented herein that supports media access control (MAC) address rotation (RCM) by generating a passcode for accessing a user defined network by one or more endpoint

devices instead of using MAC addresses for their respective device identity.
(10) In one form, a computing device, obtains a registration request for establishing a user defined network (UDN) and generates a UDN identifier and a unique passcode associated with the UDN identifier. The unique passcode enables an authentication of one or more endpoint devices to connect to the UDN. The authentication is independent of a media access control (MAC) address of a respective endpoint device. The computing device provides the UDN identifier and the unique passcode such that the UDN identifier and the unique passcode are provided to the one or more endpoint devices to connect to the UDN.
(11) In another form, a network device, obtains, from an endpoint device, a unique passcode associated with a user defined network (UDN). The network device establishes a connection for the endpoint device to a communication network. The UDN is a portion of the communication network segmented for a user of the endpoint device. The network device generates a hash of the unique passcode and provides the hash, to an authentication server. The authentication server authenticates the endpoint device onto the UDN based on the hash without using a media access control (MAC) address of the endpoint device. The network device further obtains, from the authentication server, a UDN identifier associated with the hash and connects the endpoint device to the UDN based on the UDN identifier.
EXAMPLE EMBODIMENTS
(12) The UDN uses MAC address of the respective device as a unique device identifier. For example, the user logs into a mobile application, a web portal, or other mechanism (referred to as a UDN client portal), using credentials provided by an enterprise network (referred to as a shared communication network). The user may then manually input MAC address of their devices and invite other users to be a part of their private network segment. A UDN cloud service then assigns the registered devices of the user to their personal network and stores the MAC addresses with the unique identity of the private network segment (referred to as a UDN identifier (UDN-ID)).
(13) Network devices such as a wireless local access network (LAN) controller (WLC) and access points (APs), use the UDN-ID to enforce traffic containment for the traffic generated by the registered endpoint devices. By performing MAC address filtering, traffic containment policies are enforced. For example, unicast traffic between two different personal networks is blocked.
(14) The UDN functions are based on the uniqueness of the MAC address for the respective endpoint device. Rotating/randomized changing of the MAC addresses (RCM) support on the endpoint devices creates challenges for the UDN functions because the uniqueness of the MAC address as an identity of the device is lost. In other words, each time a MAC address rotation occurs (an RCM event), the endpoint device needs to be re-authenticated onto the UDN.
(15) Further, some operating systems of the endpoint devices disable applications from fetching the MAC address through an application programming interface (API). As such, using the MAC address as the unique identity of the endpoint device is becoming increasingly more difficult.
(16) In one or more example embodiments, techniques are provided in which a unique passcode is generated and is associated with the UDN-ID. The unique passcode is used by various endpoint devices instead of their MAC address. In these techniques, a unique passcode is generated per UDN-ID and is used for authenticating user's endpoint devices onto a personal network segment (UDN) irrespective of their MAC addresses and/or MAC addresses changes. These techniques are also applicable to endpoint devices that use pre-shared key (PSK), identity PSK (iPSK), simultaneous authentication of equals (SAE) based authentication in the UDN. These techniques adapt to the changing MAC addresses of the endpoint devices when the respective endpoint device on-boards onto a network in the context of the UDN using a PSK based authentication mechanism. Because the unique passcode is assigned to each private room in addition to the UDN-ID, unique identity of the respective endpoint device is no longer needed for authenticating the endpoint device onto the UDN.
(17) While example embodiments describe generating one UDN (private room, personal network,

etc.), it is understood that a number of UDNs and private rooms being generated are not limited to one and may vary based on a particular deployment of the network and use case scenario.

(18) FIG. **1** is a block diagram illustrating a system **100** configured to authenticate one or more endpoint devices onto the UDN using a unique passcode, according to an example embodiment.

(19) The system **100** includes a shared communication network **110**, one or more access points depicted as an access point (AP) **120**, wireless local area network (LAN) controller (WLC) **130**, an authentication service **140**, a network management service **150**, a UDN service **160**, a UDN client portal **170** that communicates with one or more endpoint devices that are depicted as an endpoint device **180**.

(20) This is only an example of the system **100**, and the number and types of entities may vary based on a particular deployment and use case scenario, such as the type of service being provided and network structures. For example, while the system **100** includes the AP **120**, other network devices may be present in the system **100**. The network devices may include, but are not limited to switches, virtual routers, leaf nodes, spine nodes, etc.

(21) In various example embodiments, the entities of the system **100** (endpoint device **180**, UDN service **160**, network management service **150**, authentication service **140**, the WLC **130**, and the AP **120**) may each include a network interface, at least one processor, and a memory. Each entity may be an apparatus or any programmable electronic device capable of executing computer readable program instructions. The network interface may include one or more network interface cards (having one or more ports) that enable components of the entity to send and receive packets or data over the network(s), such as a local area network (LAN) or a wide area network (WAN), and/or wireless access networks such as the shared communication network **110**. Each entity may include internal and external hardware components such as those depicted and described in further detail in FIG. **7**. In one example, at least some of these entities may be embodied as virtual devices with functionality distributed over a number of hardware devices, such as virtual APs, switches, routers, servers, etc.

(22) The endpoint device **180** is any suitable device configured to initiate a flow in the system **100**, such as data source device and/or data sink device. For example, the endpoint device **180** may include a computer, an enterprise device, an appliance, an Internet of Things (IoT) device, a Personal Digital Assistant (PDA), a laptop or electronic notebook, a smartphone, a tablet, and/or any other device and/or combination of devices, components, elements, and/or objects capable of initiating voice, audio, video, media, or data exchanges within the system **100**. The endpoint device **180** may also include any suitable interface to a human user such as a microphone, a display, a keyboard, or other terminal equipment. The endpoint device **180** may be configured with appropriate hardware (e.g., processor(s), memory element(s), antennas and/or antenna arrays, baseband processors (modems), and/or the like such as those depicted and described in further detail in FIG. **7**), software, logic, and/or the like to facilitate respective Over-the-Air (OTA) interfaces for accessing/connecting to the AP **120** and sending or receiving packets.

(23) The endpoint device **180** is configured to connect the user to the UDN client portal **170** such as a mobile application executing on the endpoint device **180** or a web portal accessible via the endpoint device **180**. Using the UDN client portal **170**, the user registers one or more of their endpoint devices for the UDN **112** (private room) and invites other users to join the private room, as detailed below.

(24) To onboard onto the shared communication network **110**, the endpoint device **180** is connected to (establishes an association with) the AP **120**. The endpoint device **180** and the AP **120** may represent a wireless infrastructure that provides Wireless Local Area Network (WLAN) coverage for a specific geographic area/location. For example, wireless infrastructure may serve an airport, a shopping mall, a train station, a venue, etc. The endpoint device **180** and the AP **120** may use various wireless access network protocols, such as the WiFi® wireless technology, to send and receive various packets. In one example, the endpoint device **180** may be configured to connect to a

WLAN (e.g., through the AP **120**), and initially, may be part of the shared communication network **110** (e.g., a Wi-Fi® network offered in a corporate, enterprise, or dorm room environment).
(25) The AP **120** may be WLAN APs configured with appropriate hardware (e.g., processor(s), memory element(s), antennas and/or antenna arrays, baseband processors (modems), and/or the like), software, logic, and/or the like to provide OTA coverage for a WLAN access network (e.g., Wi-Fi®). In various example embodiments, the AP **120** may be implemented as Wi-Fi access point (AP) and/or the like. The AP **120** may be configured with appropriate hardware (e.g., processor(s), memory element(s), antennas and/or antenna arrays, baseband processors (modems), and/or the like), software, logic, and/or the like to facilitate respective OTA interfaces for accessing/connecting to the endpoint device **180** (to send and receive packets) and for communicating with the network management service **150** (to send and receive packets) and the UDN service **160**. The AP **120** may be managed or controlled by the network management service **150** and/or the WLC **130**. The AP **120** is connected to the WLC **130** via the LAN/WAN to send and receive data or packets.
(26) The WLC **130** may be a control plane entity that provides or is responsible for WLAN functions such as WLAN-based access authentication services, authorization services, intrusion prevention, Radio Frequency (RF) management, and/or the like to facilitate the endpoint device **180** connectivity via the AP **120**. In one form, the WLC **130** may be a software process running on one or more servers in a cloud (on any server in a datacenter or at any location with Internet connectivity). The WLC **130** is configured with appropriate hardware (e.g., processor(s), memory element(s), and/or the like such as those depicted and described in further detail in FIG. **7**), software, logic, and/or the like.
(27) The AP **120** and the WLC **130** are network devices that enable traffic containment within the UDN **112** using a UDN-ID and an associated unique passcode. The UDN **112** is a personal, secure network, a private network, a network segment dedicated to the user, or a private room within the shared communication network **110**. The user may create the UDN **112** and add the endpoint devices (e.g., the endpoint device **180**) thereto using the UDN client portal **170**, such as a mobile application downloadable to the endpoint device **180**. The UDN client portal **170** may cause the endpoint device **180** to scan the shared communication network **110** for shareable endpoint devices and display an indication of such endpoint devices in a form of a list, for example. The user selects the one or more endpoint devices to add to UDN **112**. Hence, the UDN **112** includes the registered devices of the user.
(28) Further, the AP **120** and/or the WLC **130** may perform MAC address rotation for the endpoint device **180**, at a preset time interval (each epoch), and/or for each endpoint device **180** associated with it (respective S SID). In one form, the MAC address rotation may be performed at a request of the endpoint device **180** and/or at the direction of the network management service **150**. MAC address rotation involves assigning one or more new MAC addresses to the respective endpoint device **180**. The MAC address rotation is performed by the end-point device **180** itself but may be initiated by the AP **120** and/or WLC **130**.
(29) In one or more example embodiments, the WLC **130** communicates with an authentication service **140** to perform authentication of the endpoint device **180**. The WLC facilitates and controls connectivity of the endpoint device **180** via the AP **120**. The authentication service **140**, embodied by one or more hardware computing devices, may include at least part of one or more of a digital network architecture, identity services engine (ISE), Policy Control Function (PCF), an authentication, authorization, and accounting (AAA) service, etc. The authentication service **140** communicates with the network management service **150** to authenticate the user onto the shared communication network **110** i.e., the UDN **112** of the shared communication network **110**.
(30) The network management service **150** may be a management device(s) or software process associated with wireless infrastructure. In one form, the network management service **150** may be an on-premise device(s) that connects with the UDN service **160** to provision the shared

communication network **110** and to provide visibility through telemetry and assurance data. In another form, the network management service **150** may be a software process running on one or more servers in a cloud (on any server in a datacenter or at any location with Internet connectivity). The network management service **150** is configured with appropriate hardware (e.g., processor(s), memory element(s), and/or the like such as those depicted and described in further detail in FIG. **7**), software, logic, and/or the like.

(31) The network management service **150** communicates with the UDN service **160** and the authentication service **140** to register the endpoint device **180** onto the UDN **112** and/or to onboard the endpoint device **180** onto the shared communication network **110**. In one example, the network management service **150** may communicate directly with the WLC **130** using a Remote Authentication Dial-In User Service (RADIUS) network protocol. While the authentication service **140** and the network management service **150** are depicted as separate entities, in one or more example embodiments, these entities may be embodied into an integrated service.

(32) The UDN service **160** renders the endpoint device **180** undiscoverable to other endpoint devices that are outside the UDN **112** as defined by the UDN-ID. This may be useful for protecting the security of the endpoint device **180**. The user selects one or more endpoint devices such as the endpoint device **180** to add to the UDN **112**. The UDN service **160** generates the UDN-ID and the unique passcode and at **190**, provides the UDN-ID and passcode to the UDN client portal **170**, which then disseminates the UDN-ID and passcode to the registered endpoint devices. In other words, the UDN client portal provisions the registered endpoint devices using a device provisioning protocol to connect to the UDN.

(33) The UDN service **160** is further configured to generate a hash of the passcode. At **192**, the UDN-ID and the hashed passcode are provided to the network management service **150**. The network management service **150** is configured to communicate with the authentication service **140** and optionally, the WLC **130** to provide the information about the UDN **112** (the UDN-ID and the hashed passcode). Instead of the MAC addresses being used as an identification of the registered endpoint devices, the passcode is used as the identity of the endpoint devices i.e., to indicate that these endpoint devices are members of the UDN **112**, as explained in further detail below.

(34) With continued reference to FIG. **1**, reference is now made to FIGS. **2-4**, which illustrate various UDN processes, according to one or more example embodiments. The UDN processes involve various network entities. For example, various network entities include one or more endpoint devices **210***a-n* (EPs **210***a-n*) that are registered for the UDN **112** of FIG. **1**. The network entities further include a mobile application (mobile app) **212**, which is an example of the UDN client portal **170** of FIG. **1** and a cloud-UDN **214**, which is an example of the UDN service **160** of FIG. **1**. The network entities further include an authentication server **216**. The authentication server **216** is an example of integrated network management service **150** and authentication service **140**. The network entities further include the WLC **218** such as the WLC **130** of FIG. **1**.

(35) The notation “a-n” denotes that a number is not limited, can vary widely, and depends on a particular use case scenario, and need not be the same, in number, for the endpoint devices, APs, etc.

(36) Specifically, FIG. **2** is a flow diagram illustrating a UDN registration process **200** in which a unique passcode is generated for each private room in addition to the unique UDN-ID, according to an example embodiment.

(37) The UDN registration process **200** involves at **220**, the mobile app **212** registers the device (one of the EPs **210***a-n*) for the UDN **112**. Specifically, the device on which the mobile app **212** is executing, sends a registration request to the cloud-UDN **214**.

(38) At **222**, the cloud-UDN **214** generates a unique UDN-ID for the UDN **112** and a unique passcode that is associated with the private room (the UDN **112**). The unique passcode is a unique value that is associated with and linked to the UDN-ID. As such, all EPs **210***a-n* that are to be part of the UDN **112** use this unique passcode (instead of their MAC addresses) to join the UDN **112**.

The passcode enables the EPs **210***a-n* to join the UDN **112** regardless of their MAC addresses and/or MAC address changes. The unique passcode may be a passphrase or a pre-shared key (PSK). The passphrase and/or the PSK is generated as a function of the UDN-ID e.g., function (UDN-ID). The passcode is a unique string of characters that may be used to generate encryption keys. In other words, while the MAC address is included in the packet, the MAC address is not used for an assignment of an EP to the UDN **112**. In one example embodiment, when the UDN-ID is generated, the UDN-ID is used as input to a function to generate the passcode.
(39) At **224**, the generated UDN information is pushed from the cloud-UDN **214** to the mobile app **212**. The UDN information includes the generated UDN-ID and unique passcode.
(40) At **226**, the mobile app **212** pushes the generated passcode (passphrase or PSK) to the EPs **210***a-n* using various mechanisms such as a device provisioning protocol (DPP). For example, PSK-ID (one example of the passcode) is provided to PSK-based endpoint devices that are to be part of the private room/the UDN **112**. That is, the mobile app **212** provisions the EPs **210***a-n* to connect to the UDN **112**. The UDN **112** is a portion of the common communication network segmented for the user.
(41) Additionally, at **228**, the cloud-UDN **214** hashes the unique passcode of the private room/the UDN **112**. For example, the cloud-UDN **214** generates a hash over the auto-generated PSK-ID for each private room or generates a hash from the UDN-ID and PSK-ID binding. At **230**, the cloud-UDN **214** communicates UDN information to the authentication server **216**. For example, the UDN information includes a hash of the PSK-ID, the UDN-ID, and other information for the private room/the UDN **112**. The other information may include user related information (user profile). At **232**, the authentication server **216** stores the UDN-ID and the hash of the unique passcode (e.g., the PSK-hash) for the private room/the UDN **112** in its datastore. The UDN-ID and the hash are stored in association with one another along with other information about the UDN **112** such as traffic constraint policies, user profile, etc.
(42) When a user wans to invite another trusted user to join the UDN **112**, the user, using the mobile app **212**, generates an invitation to join for the other trusted user. The other trusted user obtains the invitation via the cloud-UDN **214**. If the trusted user accepts the invitation, the cloud-UDN **214** pushes the unique passcode (PSK-ID) to the endpoint device of the trusted user. The endpoint device of the trusted user then connects to the UDN using the unique passcode, as described below.
(43) FIG. **3** is a flow diagram illustrating a process **300** of onboarding a first endpoint device **210***a* onto a UDN, according to an example embodiment. The first endpoint device **210***a* may be a device of the user that is executing the mobile app **212**, may be another endpoint device of the user, or may be an endpoint device of the trusted user that was invited to join the UDN.
(44) The onboarding process **300** includes at **302**, the first endpoint device **210***a* joining a wireless network. In one example, the first endpoint device **210***a* is PSK enabled client device and the wireless network is an example of the shared communication network **110** of FIG. **1** such as a WiFi® network offered in a corporate enterprise or shared environment (dorm room environment, multi-dwelling building units, etc.). The AP **120** of FIG. **1** and/or the WLC **218** may use the MAC address of the first endpoint device **210***a* to authenticate the first endpoint device **210***a* onto the common wireless network. The first endpoint device **210***a* further uses the unique passcode such as the auto-generated PSK, to join the UDN **112** of FIGS. **1** and **2** of the common wireless network.
(45) Specifically, in response to receiving the unique passcode from the first endpoint device **210***a* (via the AP **120** of FIG. **1**), at **304**, the WLC **218** generates a hash of the unique passcode. That is, the WLC **218** generates a PSK-hash using the same algorithm that is used by the cloud-UDN **214** at **228** of FIG. **2**. At **306**, the WLC **218** sends the generated hash, as a vendor payload in the RADIUS message, for example, to the authentication server **216** along with the MAC address of the first endpoint device **210***a*.
(46) The authentication server **216** authenticates the first endpoint device **210***a*. At **308**, based on

the PSK-hash, the authentication server **216** further determines the UDN-ID that corresponds the PSK-hash (independent of the MAC address). That is, the authentication server **216** compares the PSK-hash with hashes stored in its datastore/database to find a match. If a match is found, the authentication server **216** retrieves a corresponding UDN-ID that is mapped to the PSK-hash. In other words, regardless of the MAC address of the first endpoint device **210***a* (even after the MAC address of the first endpoint device **210***a* is rotated), the authentication server **216** determines the UDN-ID that is associated with the PSK-hash. At **310**, in response to a successful authentication of the first endpoint device **210***a*, the authentication server **216** transmits to the WLC **218**, an authentication success message that includes the UDN-ID along with other attributes. The authentication success message is provided using defined RADIUS attributes, for example.

(47) In one example, the WLC **218** and the authentication server **216** communicate with each other using the RADIUS protocol. The WLC **218** generates RADIUS request messages with information required for an authentication of the user and/or the first endpoint device **210***a*. The information may be provided in the vendor payload. The authentication server **216** provides RADIUS responses that include the UDN identifier as one of a plurality of pre-defined RADIUS attributes.

(48) At **312**, the WLC **218** successfully onboards the first endpoint device **210***a* and enforces required traffic containment on the first endpoint device **210***a*. That is, the WLC **218** puts the first endpoint device **210***a* into the private room based on the UDN-ID. The WLC **218** then applies one or more traffic constraint policies associated with the UDN on one or more network packets obtained for the first endpoint device **210***a*.

(49) FIG. **4** is a flow diagram illustrating a process **400** of updating UDN information at a UDN client portal based on a MAC address rotation, according to an example embodiment.

(50) At **402**, the first endpoint device **210***a* joins another SSID or rotates the MAC address. The new MAC address is then on-boarded onto the common wireless network. Specifically, at **404**, the new MAC address is propagated from the first endpoint device **210***a* to the WLC **218**. The WLC **218** updates its records and associates the new MAC address with the common wireless network. Additionally, the WLC **218** generates a hash of the passcode associated with the private room of the first endpoint device **210***a* and at **406**, the WLC **218** provides the new MAC address and the hashed passcode to the authentication server **216**. The authentication server **216** sends access-accept to the WLC **218** to on-board the first endpoint device **210***a*.

(51) The authentication server **216** determines the UDN-ID that is associated with the hashed passcode and at **408**, the new MAC address along with the retrieved UDN-ID is propagated from the authentication server **216** to the cloud-UDN **214**. At **410**, the new MAC address along with the UDN-ID is propagated from the cloud-UDN **214** to the mobile app **212**. At **412**, the mobile app **212** adds the new MAC address to the private room identified by the UDN-ID. The mobile app **212** may further show or display the new MAC address that has joined the private room.

(52) To ensure that the old MAC address is removed from the private room/the UDN (after the MAC address has been rotated), various mechanisms may be used such as a Dynamic Host Configuration Protocol (DHCP) scavenger timers.

(53) For example, at **414**, the WLC **218** deletes the old MAC address in response to MAC address rotation and/or after an idle time period. In response thereto, at **416**, MAC address event is generated and provided to the authentication server **216**. That is, the event includes the deleted MAC address and a corresponding hashed passcode. The authentication server **216** determines the UDN-ID based on the hashed passcode and at **418**, notifies the cloud-UDN **214** of the deleted MAC address associated with the UDN-ID. At **420**, the cloud-UDN **214** provides the deleted MAC address with the UDN-ID to the mobile app **212**. At **422**, the mobile app **212** removes the corresponding entry from the private room/UDN i.e., the old MAC address is removed from the UDN. That is, the mobile app **212** updates the MAC address information by adding new MAC address of the first endpoint device **210***a* and deleting the old MAC address of the first endpoint device **210***a*.

(54) MAC address information is propagated from the WLC **218** to the mobile app **212** such that MAC addresses of the endpoint devices are managed by the mobile app **212** and are not required by the cloud-UDN **214**. That is, any RCM methods coupled with a PSK based authenticated endpoint device are supported using the passcode instead of the MAC address authentication. Additionally, when an endpoint device joins another UDN enabled SSID after the initial on-boarding, the MAC address is not needed, and a sharable passcode is used instead. In other words, the endpoint device need not re-authenticate onto the UDN with the new MAC address and may skip the secondary authentication. Even if the MAC address has changed, the endpoint device remains connected to the UDN. Since UDN authentication is independent of the identity of the endpoint device, re-association with the UDN is no longer needed.
(55) RCM and the lack of stable identifiers creates challenges with establishing policies for PSK networks. However, the technique presented herein use the UDN-ID to generate a unique passcode per UDN-ID that enables an authentication of the endpoint devices with the network irrespective of their identities (e.g., MAC addresses). Rotating or randomizing identities of the endpoint devices does not influence their connections to the UDN.
(56) FIG. **5** is a flow diagram illustrating a method **500** of generating a unique passcode that is used instead of a MAC address to connect an endpoint device to the UDN, according to an example embodiment. The method **500** may be performed by a computing device such as one or more servers of the UDN service **160** of FIG. **1** or the cloud-UDN **214** of FIGS. **2** and **4**.
(57) The method **500** involves, at **502**, obtaining a registration request for establishing a user defined network (UDN) and at **504**, generating a UDN identifier and a unique passcode associated with the UDN identifier. The unique passcode enables an authentication of one or more endpoint devices to connect to the UDN. The authentication is independent of the MAC address of a respective endpoint device.
(58) The method **500** further involves at **506**, providing the UDN identifier and the unique passcode such that the UDN identifier and the unique passcode are provided to the one or more endpoint devices to connect to the UDN.
(59) In one instance, the registration request, obtained at **502**, may be obtained from a mobile application that is executing on the respective endpoint device of the one or more endpoint devices. Further, the UDN identifier and the unique passcode may be provided to the mobile application, which provisions the one or more endpoint devices using a device provisioning protocol to connect to the UDN using the UDN identifier and the unique passcode.
(60) In one form, the method **500** may further involve obtaining, from an authentication server, a new MAC address of the respective endpoint device. The new MAC address may be generated based on performing a MAC address rotation. The method **500** may further involve providing, to the mobile application, the new MAC address and the UDN identifier such that the mobile application updates the MAC address of the respective endpoint device associated with the UDN.
(61) In one or more example embodiments, the respective endpoint device may be authenticated to connect to the UDN using the unique passcode instead of the MAC address.
(62) In one form, the method **500** may further involve generating a hash of the unique passcode and providing, to an authentication server, UDN information including the hash and the UDN identifier.
(63) In one instance, the operation **504** of generating the unique passcode may involve generating a passphrase as a function of the UDN identifier.
(64) In another instance, the operation **504** of generating the unique passcode may involve generating a pre-shared key as a function of the UDN identifier.
(65) FIG. **6** is a flowchart illustrating a method of **600** of connecting an endpoint device to the UDN based on a passcode provided by the endpoint device, according to an example embodiment. The method **600** may be performed by a network device, such as the WLC **130** of FIG. **1** or the WLC **218** of FIGS. **3** and **4**.
(66) The method **600** involves at **602**, obtaining, from an endpoint device, a unique passcode

associated with a user defined network (UDN). A connection for the endpoint device to a communication network is established and the UDN is a portion of the communication network segmented for a user of the endpoint device.
(67) The method **600** further involves at **604**, generating a hash of the unique passcode.
(68) The method **600** further involves at **606**, providing the hash to an authentication server. The authentication server authenticates the endpoint device onto the UDN based on the hash without using a media access control (MAC) address of the endpoint device.
(69) The method **600** further involves at **608**, obtaining, from the authentication server, a UDN identifier associated with the hash and at **610**, connecting the endpoint device to the UDN based on the UDN identifier.
(70) In one or more example embodiments, the unique passcode and the UDN identifier may be generated by a UDN service. The hash of the unique passcode is associated with the UDN identifier and is stored at the authentication server.
(71) In one instance, the operation **606** of providing the hash may involve providing a Remote Authentication Dial-In User Service (RADIUS) message that includes the hash in a vendor payload.
(72) In one form, the operation **608** of obtaining the UDN identifier may involve obtaining a RADIUS message having the UDN identifier as one of a plurality of pre-defined RADIUS attributes. The UDN identifier may be matched with the hash by the authentication server.
(73) In one or more example embodiments, the operation **610** of connecting the endpoint device to the UDN may involve applying one or more traffic constraint policies associated with the UDN to one or more network packets obtained from the endpoint device.
(74) In one instance, the unique passcode may be a passphrase generated as a function of the UDN identifier.
(75) In another instance, the unique passcode may be a pre-shared key generated as a function of the UDN identifier.
(76) FIG. **7** is a hardware block diagram of a computing device **700** that may perform functions associated with any combination of operations in connection with the techniques depicted in FIGS. **1-6**, according to various example embodiments, including, but not limited to, operations of the one or more endpoint devices such as endpoint device **180** of FIG. **1**, EPs **210***a-n* of FIG. **2**, or the first endpoint device **210***a* of FIGS. **3** and **4**. Further the computing device **700** may be representative of the WLC **130**, the authentication service **140**, the network management service **150**, or the UDN service **160** that are shown in FIG. **1**. Further, the computing device **700** may be representative of the cloud-UDN **214**, the authentication server **216**, or the WLC **218** of FIGS. **2-4**. It should be appreciated that FIG. **7** provides only an illustration of one example embodiment and does not imply any limitations with regard to the environments in which different example embodiments may be implemented. Many modifications to the depicted environment may be made.
(77) In at least one embodiment, computing device **700** may include one or more processor(s) **702**, one or more memory element(s) **704**, storage **706**, a bus **708**, one or more network processor unit(s) **710** interconnected with one or more network input/output (I/O) interface(s) **712**, one or more I/O interface(s) **714**, and control logic **720**. In various embodiments, instructions associated with logic for computing device **700** can overlap in any manner and are not limited to the specific allocation of instructions and/or operations described herein.
(78) In at least one embodiment, processor(s) **702** is/are at least one hardware processor configured to execute various tasks, operations and/or functions for computing device **700** as described herein according to software and/or instructions configured for computing device **700**. Processor(s) **702** (e.g., a hardware processor) can execute any type of instructions associated with data to achieve the operations detailed herein. In one example, processor(s) **702** can transform an element or an article (e.g., data, information) from one state or thing to another state or thing. Any of potential processing elements, microprocessors, digital signal processor, baseband signal processor, modem,

PHY, controllers, systems, managers, logic, and/or machines described herein can be construed as being encompassed within the broad term ‘processor’.

(79) In at least one embodiment, one or more memory element(s) **704** and/or storage **706** is/are configured to store data, information, software, and/or instructions associated with computing device **700**, and/or logic configured for memory element(s) **704** and/or storage **706**. For example, any logic described herein (e.g., control logic **720**) can, in various embodiments, be stored for computing device **700** using any combination of memory element(s) **704** and/or storage **706**. Note that in some embodiments, storage **706** can be consolidated with one or more memory elements **704** (or vice versa), or can overlap/exist in any other suitable manner.

(80) In at least one embodiment, bus **708** can be configured as an interface that enables one or more elements of computing device **700** to communicate in order to exchange information and/or data. Bus **708** can be implemented with any architecture designed for passing control, data and/or information between processors, memory elements/storage, peripheral devices, and/or any other hardware and/or software components that may be configured for computing device **700**. In at least one embodiment, bus **708** may be implemented as a fast kernel-hosted interconnect, potentially using shared memory between processes (e.g., logic), which can enable efficient communication paths between the processes.

(81) In various embodiments, network processor unit(s) **710** may enable communication between computing device **700** and other systems, entities, etc., via network I/O interface(s) **712** to facilitate operations discussed for various embodiments described herein. In various embodiments, network processor unit(s) **710** can be configured as a combination of hardware and/or software, such as one or more Ethernet driver(s) and/or controller(s) or interface cards, Fibre Channel (e.g., optical) driver(s) and/or controller(s), and/or other similar network interface driver(s) and/or controller(s) now known or hereafter developed to enable communications between computing device **700** and other systems, entities, etc. to facilitate operations for various embodiments described herein. In various embodiments, network I/O interface(s) **712** can be configured as one or more Ethernet port(s), Fibre Channel ports, and/or any other I/O port(s) now known or hereafter developed. Thus, the network processor unit(s) **710** and/or network I/O interface(s) **712** may include suitable interfaces for receiving, transmitting, and/or otherwise communicating data and/or information in a network environment.

(82) I/O interface(s) **714** allow for input and output of data and/or information with other entities that may be connected to computing device **700**. For example, I/O interface(s) **714** may provide a connection to external devices such as a keyboard, keypad, a touch screen, and/or any other suitable input device now known or hereafter developed. In some instances, external devices can also include portable computer readable (non-transitory) storage media such as database systems, thumb drives, portable optical or magnetic disks, and memory cards. In still some instances, external devices can be a mechanism to display data to a user, such as, for example, a computer monitor **716**, a display screen, or the like.

(83) In various embodiments, control logic **720** can include instructions that, when executed, cause processor(s) **702** to perform operations, which can include, but not be limited to, providing overall control operations of computing device; interacting with other entities, systems, etc. described herein; maintaining and/or interacting with stored data, information, parameters, etc. (e.g., memory element(s), storage, data structures, databases, tables, etc.); combinations thereof; and/or the like to facilitate various operations for embodiments described herein.

(84) In another example embodiment, an apparatus is provided. The apparatus includes a network interface to receive and send packets in a network and a processor. The processor is configured to perform various operations including obtaining, from the network interface, a registration request for establishing a user defined network (UDN) and generating a UDN identifier and a unique passcode associated with the UDN identifier. The unique passcode enables an authentication of one or more endpoint devices to connect to the UDN and the authentication is independent of a media

access control (MAC) address of a respective endpoint device. The operations further include providing, to the network interface, the UDN identifier and the unique passcode such that the UDN identifier and the unique passcode are provided to the one or more endpoint devices to connect to the UDN.

(85) In yet another example embodiment, an apparatus is provided. The apparatus includes a network interface configured to receive and send packets in a network and a processor. The processor is configured to perform various operations. The operations include obtaining, from an endpoint device, a unique passcode associated with a user defined network (UDN). The apparatus establishes a connection to a communication network for the endpoint device. The UDN is a portion of the communication network segmented for a user of the endpoint device. The operations further include generating a hash of the unique passcode and providing the hash to an authentication server. The authentication server authenticates the endpoint device onto the UDN based on the hash without using a media access control (MAC) address of the endpoint device. The operations further involve obtaining, from the authentication server, a UDN identifier associated with the hash and connecting the endpoint device to the UDN based on the UDN identifier.

(86) In yet another example embodiment, one or more non-transitory computer readable storage media encoded with instructions are provided. When the media is executed by a processor, the instructions cause the processor to execute a method that involves obtaining a registration request for establishing a user defined network (UDN) and generating a UDN identifier and a unique passcode associated with the UDN identifier. The unique passcode enables an authentication of one or more endpoint devices to connect to the UDN. The authentication is independent of a media access control (MAC) address of a respective endpoint device. The method further involves providing the UDN identifier and the unique passcode such that the UDN identifier and the unique passcode are provided to the one or more endpoint devices to connect to the UDN.

(87) In yet another example embodiment, one or more non-transitory computer readable storage media encoded with instructions are provided. When the media is executed by a processor, the instructions cause the processor to execute another method that involves obtaining, from an endpoint device, a unique passcode associated with a user defined network (UDN). A connection to a communication network is established for the endpoint deice. The UDN is a portion of the communication network segmented for a user of the endpoint device. The method further involves generating a hash of the unique passcode and providing the hash to an authentication server. The authentication server authenticates the endpoint device onto the UDN based on the hash without using a media access control (MAC) address of the endpoint device. The method further involves obtaining, from the authentication server, a UDN identifier associated with the hash and connecting the endpoint device to the UDN based on the UDN identifier.

(88) In yet another example embodiment, a system is provided that includes the devices and operations explained above with reference to FIGS. **1-7**.

(89) The programs described herein (e.g., control logic **720**) may be identified based upon the application(s) for which they are implemented in a specific embodiment. However, it should be appreciated that any particular program nomenclature herein is used merely for convenience, and thus the embodiments herein should not be limited to use(s) solely described in any specific application(s) identified and/or implied by such nomenclature.

(90) In various embodiments, entities as described herein may store data/information in any suitable volatile and/or non-volatile memory item (e.g., magnetic hard disk drive, solid state hard drive, semiconductor storage device, random access memory (RAM), read only memory (ROM), erasable programmable read only memory (EPROM), application specific integrated circuit (ASIC), etc.), software, logic (fixed logic, hardware logic, programmable logic, analog logic, digital logic), hardware, and/or in any other suitable component, device, element, and/or object as may be appropriate. Any of the memory items discussed herein should be construed as being encompassed within the broad term ‘memory element’. Data/information being tracked and/or sent

to one or more entities as discussed herein could be provided in any database, table, register, list, cache, storage, and/or storage structure: all of which can be referenced at any suitable timeframe. Any such storage options may also be included within the broad term ‘memory element’ as used herein.
(91) Note that in certain example implementations, operations as set forth herein may be implemented by logic encoded in one or more tangible media that is capable of storing instructions and/or digital information and may be inclusive of non-transitory tangible media and/or non-transitory computer readable storage media (e.g., embedded logic provided in: an ASIC, digital signal processing (DSP) instructions, software [potentially inclusive of object code and source code], etc.) for execution by one or more processor(s), and/or other similar machine, etc. Generally, the storage **706** and/or memory elements(s) **704** can store data, software, code, instructions (e.g., processor instructions), logic, parameters, combinations thereof, and/or the like used for operations described herein. This includes the storage **706** and/or memory elements(s) **704** being able to store data, software, code, instructions (e.g., processor instructions), logic, parameters, combinations thereof, or the like that are executed to carry out operations in accordance with teachings of the present disclosure.
(92) In some instances, software of the present embodiments may be available via a non-transitory computer useable medium (e.g., magnetic or optical mediums, magneto-optic mediums, CD-ROM, DVD, memory devices, etc.) of a stationary or portable program product apparatus, downloadable file(s), file wrapper(s), object(s), package(s), container(s), and/or the like. In some instances, non-transitory computer readable storage media may also be removable. For example, a removable hard drive may be used for memory/storage in some implementations. Other examples may include optical and magnetic disks, thumb drives, and smart cards that can be inserted and/or otherwise connected to a computing device for transfer onto another computer readable storage medium.
(93) Embodiments described herein may include one or more networks, which can represent a series of points and/or network elements of interconnected communication paths for receiving and/or transmitting messages (e.g., packets of information) that propagate through the one or more networks. These network elements offer communicative interfaces that facilitate communications between the network elements. A network can include any number of hardware and/or software elements coupled to (and in communication with) each other through a communication medium. Such networks can include, but are not limited to, any local area network (LAN), virtual LAN (VLAN), wide area network (WAN) (e.g., the Internet), software defined WAN (SD-WAN), wireless local area (WLA) access network, wireless wide area (WWA) access network, metropolitan area network (MAN), Intranet, Extranet, virtual private network (VPN), Low Power Network (LPN), Low Power Wide Area Network (LPWAN), Machine to Machine (M2M) network, Internet of Things (IoT) network, Ethernet network/switching system, any other appropriate architecture and/or system that facilitates communications in a network environment, and/or any suitable combination thereof.
(94) Networks through which communications propagate can use any suitable technologies for communications including wireless communications (e.g., 4G/5G/nG, IEEE 802.11 (e.g., Wi-Fi®/Wi-Fi6®), IEEE 802.16 (e.g., Worldwide Interoperability for Microwave Access (WiMAX)), Radio-Frequency Identification (RFID), Near Field Communication (NFC), Bluetooth™ mm.wave, Ultra-Wideband (UWB), etc.), and/or wired communications (e.g., T1 lines, T3 lines, digital subscriber lines (DSL), Ethernet, Fibre Channel, etc.). Generally, any suitable means of communications may be used such as electric, sound, light, infrared, and/or radio to facilitate communications through one or more networks in accordance with embodiments herein. Communications, interactions, operations, etc. as discussed for various embodiments described herein may be performed among entities that may directly or indirectly connected utilizing any algorithms, communication protocols, interfaces, etc. (proprietary and/or non-proprietary) that allow for the exchange of data and/or information.

(95) Communications in a network environment can be referred to herein as ‘messages’, ‘messaging’, ‘signaling’, ‘data’, ‘content’, ‘objects’, ‘requests’, ‘queries’, ‘responses’, ‘replies’, etc. which may be inclusive of packets. As referred to herein, the terms may be used in a generic sense to include packets, frames, segments, datagrams, and/or any other generic units that may be used to transmit communications in a network environment. Generally, the terms reference to a formatted unit of data that can contain control or routing information (e.g., source and destination address, source and destination port, etc.) and data, which is also sometimes referred to as a ‘payload’, ‘data payload’, and variations thereof. In some embodiments, control or routing information, management information, or the like can be included in packet fields, such as within header(s) and/or trailer(s) of packets. Internet Protocol (IP) addresses discussed herein and in the claims can include any IP version 4 (IPv4) and/or IP version 6 (IPv6) addresses.
(96) To the extent that embodiments presented herein relate to the storage of data, the embodiments may employ any number of any conventional or other databases, data stores or storage structures (e.g., files, databases, data structures, data, or other repositories, etc.) to store information.
(97) Note that in this Specification, references to various features (e.g., elements, structures, nodes, modules, components, engines, logic, steps, operations, functions, characteristics, etc.) included in ‘one embodiment’, ‘example embodiment’, ‘an embodiment’, ‘another embodiment’, ‘certain embodiments’, ‘some embodiments’, ‘various embodiments’, ‘other embodiments’, ‘alternative embodiment’, and the like are intended to mean that any such features are included in one or more embodiments of the present disclosure, but may or may not necessarily be combined in the same embodiments. Note also that a module, engine, client, controller, function, logic or the like as used herein in this Specification, can be inclusive of an executable file comprising instructions that can be understood and processed on a server, computer, processor, machine, compute node, combinations thereof, or the like and may further include library modules loaded during execution, object files, system files, hardware logic, software logic, or any other executable modules.
(98) It is also noted that the operations and steps described with reference to the preceding figures illustrate only some of the possible scenarios that may be executed by one or more entities discussed herein. Some of these operations may be deleted or removed where appropriate, or these steps may be modified or changed considerably without departing from the scope of the presented concepts. In addition, the timing and sequence of these operations may be altered considerably and still achieve the results taught in this disclosure. The preceding operational flows have been offered for purposes of example and discussion. Substantial flexibility is provided by the embodiments in that any suitable arrangements, chronologies, configurations, and timing mechanisms may be provided without departing from the teachings of the discussed concepts.
(99) As used herein, unless expressly stated to the contrary, use of the phrase ‘at least one of’, ‘one or more of’, ‘and/or’, variations thereof, or the like are open-ended expressions that are both conjunctive and disjunctive in operation for any and all possible combination of the associated listed items. For example, each of the expressions ‘at least one of X, Y and Z’, ‘at least one of X, Y or Z’, ‘one or more of X, Y and Z’, ‘one or more of X, Y or Z’ and ‘X, Y and/or Z’ can mean any of the following: 1) X, but not Y and not Z; 2) Y, but not X and not Z; 3) Z, but not X and not Y; 4) X and Y, but not Z; 5) X and Z, but not Y; 6) Y and Z, but not X; or 7) X, Y, and Z.
(100) Additionally, unless expressly stated to the contrary, the terms ‘first’, ‘second’, ‘third’, etc., are intended to distinguish the particular nouns they modify (e.g., element, condition, node, module, activity, operation, etc.). Unless expressly stated to the contrary, the use of these terms is not intended to indicate any type of order, rank, importance, temporal sequence, or hierarchy of the modified noun. For example, ‘first X’ and ‘second X’ are intended to designate two ‘X’ elements that are not necessarily limited by any order, rank, importance, temporal sequence, or hierarchy of the two elements. Further as referred to herein, ‘at least one of’ and ‘one or more of’ can be represented using the ‘(s)’ nomenclature (e.g., one or more element(s)).
(101) Each example embodiment disclosed herein has been included to present one or more

different features. However, all disclosed example embodiments are designed to work together as part of a single larger system or method. This disclosure explicitly envisions compound embodiments that combine multiple previously-discussed features in different example embodiments into a single system or method.

(102) One or more advantages described herein are not meant to suggest that any one of the embodiments described herein necessarily provides all of the described advantages or that all the embodiments of the present disclosure necessarily provide any one of the described advantages. Numerous other changes, substitutions, variations, alterations, and/or modifications may be ascertained to one skilled in the art and it is intended that the present disclosure encompass all such changes, substitutions, variations, alterations, and/or modifications as falling within the scope of the appended claims.

Claims

1. A method comprising: obtaining, by a computing device, a registration request for establishing a user defined network (UDN), wherein the UDN is a network segment within a shared communication network that is dedicated to a user of a first endpoint device to which at least one second endpoint device is invited; generating, by the computing device, a unique UDN identifier and a unique passcode associated with the unique UDN identifier, wherein the unique passcode enables an authentication of a plurality of endpoint devices including the first endpoint device and the at least one second endpoint device, to connect to the UDN and the authentication is independent of a unique identity of a respective endpoint device; providing, by the computing device, the unique UDN identifier and the unique passcode such that the unique UDN identifier and the unique passcode are for connecting the plurality of endpoint devices to the UDN; obtaining a new unique identity of the respective endpoint device based on an identity rotation in which the unique identity of the respective endpoint device is changed to the new unique identity; and providing the new unique identity and the unique UDN identifier to update the unique identity associated with the respective endpoint device connected to the UDN.
2. The method of claim 1, wherein the registration request is obtained from a mobile application that is executing on the respective endpoint device of the plurality of endpoint devices, and wherein the unique UDN identifier and the unique passcode are provided to the mobile application, which provisions the plurality of endpoint devices using a device provisioning protocol to connect to the UDN using the unique UDN identifier and the unique passcode.
3. The method of claim 2, wherein the unique identity of the respective endpoint device is a media access control (MAC) address and wherein the new unique identity is a new MAC address generated based on performing a MAC address rotation.
4. The method of claim 1, wherein the respective endpoint device is authenticated to connect to the UDN using the unique passcode instead of the unique identity.
5. The method of claim 1, further comprising: generating, by the computing device, a hash of the unique passcode; and providing, by the computing device to an authentication server, UDN information including the hash and the unique UDN identifier.
6. The method of claim 1, wherein generating the unique passcode involves generating a passphrase as a function of the unique UDN identifier.
7. The method of claim 1, wherein generating the unique passcode involves generating a pre-shared key as a function of the unique UDN identifier.
8. The method of claim 1, further comprising: establishing a first connection with a wireless access network for the respective endpoint device, which is authenticated onto the wireless access network using a media access control (MAC) address of the respective endpoint device; and establishing a second connection with the UDN, using the unique passcode instead of the MAC address.
9. The method of claim 8, wherein the second connection with the UDN is maintained without re-

association with the UDN after the identity rotation.

10. A method comprising: obtaining, by a network device from an endpoint device, a unique passcode associated with a user defined network (UDN), wherein the network device establishes a connection for the endpoint device to a communication network and wherein the UDN is a portion of the communication network segmented for a user of the endpoint device; generating, by the network device, a hash of the unique passcode; providing, by the network device, the hash, to an authentication server, wherein the authentication server authenticates the endpoint device onto the UDN based on the hash without using a unique identity of the endpoint device; obtaining, by the network device from the authentication server, a unique UDN identifier associated with the hash; connecting, by the network device, the endpoint device to the UDN based on the unique UDN identifier; obtaining a new unique identity of the endpoint device based on the endpoint device performing an identity rotation in which the unique identity of the endpoint device is changed to the new unique identity; and providing, to the authentication server, the new unique identity associated with the hash and a request to delete the unique identity associated with the hash to update the unique identity of the endpoint device associated with the UDN.

11. The method of claim 10, wherein the unique passcode and the unique UDN identifier are generated by a UDN service and wherein the hash of the unique passcode is associated with the unique UDN identifier and stored at the authentication server.

12. The method of claim 10, wherein providing the hash involves providing a Remote Authentication Dial-In User Service (RADIUS) message that includes the hash in a vendor payload.

13. The method of claim 12, wherein obtaining the unique UDN identifier involves obtaining a RADIUS message having the unique UDN identifier as one of a plurality of pre-defined RADIUS attributes, wherein the unique UDN identifier is matched with the hash by the authentication server.

14. The method of claim 10, wherein connecting the endpoint device to the UDN includes: applying, by the network device, one or more traffic constraint policies associated with the UDN to one or more network packets obtained from the endpoint device.

15. The method of claim 10, wherein the unique passcode is a passphrase generated as a function of the unique UDN identifier.

16. The method of claim 10, wherein the unique passcode is a pre-shared key generated as a function of the unique UDN identifier.

17. An apparatus comprising: a network interface to receive and send packets in a network; and a processor, wherein the processor is configured to perform operations comprising: obtaining, from the network interface, a registration request for establishing a user defined network (UDN), wherein the UDN is a network segment within a shared communication network that is dedicated to a user of the apparatus and to which at least one other endpoint device is invited; generating a unique UDN identifier and a unique passcode associated with the unique UDN identifier, wherein the unique passcode enables an authentication of a plurality of endpoint devices to connect to the UDN and the authentication is independent of a unique identity of a respective endpoint device; providing, to the network interface, the unique UDN identifier and the unique passcode such that the unique UDN identifier and the unique passcode are for connecting the plurality of endpoint devices to the UDN; obtaining, from the network interface, a new unique identity of the respective endpoint device based on an identity rotation in which the unique identity of the respective endpoint device is changed to the new unique identity; and providing the new unique identity and the unique UDN identifier to update the unique identity of the respective endpoint device connected to the UDN.

18. The apparatus of claim 17, wherein the registration request is obtained from a mobile application that is executing on the respective endpoint device, and wherein the unique UDN identifier and the unique passcode are provided to the mobile application, which provisions the plurality of endpoint devices using a device provisioning protocol to connect to the UDN using the

unique UDN identifier and the unique passcode.
19. The apparatus of claim 18, wherein the unique identity of the respective endpoint device is a media access control (MAC) address and wherein the new unique identity is a new MAC address generated based on performing a MAC address rotation.
20. The apparatus of claim 17, wherein the respective endpoint device is authenticated to connect to the UDN using the unique passcode instead of the unique identity.